

Program 5:

Project Structure:

Pr5

->Dockerfile

->docker-compose.yaml

Step 1: Create docker-compose.yaml

version: '3.8'

services:

jenkins:

build: .

container_name: jenkins-ci

ports:

- "8080:8080"

- "50000:50000"

volumes:

- jenkins_home:/var/jenkins_home

- /var/run/docker.sock:/var/run/docker.sock

volumes:

jenkins_home:

Step 2: Create Dockerfile

FROM jenkins/jenkins:its

USER root

*RUN apt-get update *

*&& apt-get install -y docker.io *

*&& rm -rf /var/lib/apt/lists/**

Add 'jenkins' user to the 'docker' group

RUN usermod -aG docker jenkins

USER jenkins

Step 3: Build and run jenkins

```
—(singh@ singh)-[~/pr5]
$ sudo docker compose build
WARN[0000] /home/singh/pr5/docker-compose.yml: the attribute `version` is not supported
[+] Building 31.0s (8/8) FINISHED
=> [internal] load local bake definitions
=> => reading from stdin 479B
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 138B
=> [internal] load metadata for docker.io/jenkins/jenkins:lts
=> [internal] load .dockerignore
=> => transferring context: 2B
=> CACHED [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:7b1c3782
=> [2/2] RUN apt-get update && apt-get install -y docker.io
=> exporting to image
=> => exporting layers
=> => writing image sha256:5ddf84919491943f1a4bbe2efc7d534c910136b
=> => naming to docker.io/library/pr5-jenkins
=> resolving provenance for metadata file
[+] Building 1/1
✓ pr5-jenkins Built
```

```
—(singh@ singh)-[~/pr5]
$ sudo docker compose up
WARN[0000] /home/singh/pr5/docker-compose.yml: the attribute `version` is not supported
[+] Running 3/3
✓ Network pr5_default Created
✓ Volume pr5_jenkins_home Created
✓ Container jenkins-ci Created
Attaching to jenkins-ci
jenkins-ci | Running from: /usr/share/jenkins/jenkins.war
jenkins-ci | webroot: /var/jenkins_home/war
jenkins-ci | 2025-11-21 09:33:11.846+0000 [id=1] INFO winston
jenkins-ci | 2025-11-21 09:33:12.403+0000 [id=1] WARNING o.e.j.e
jenkins-ci | 2025-11-21 09:33:12.434+0000 [id=1] INFO org.ecl
jenkins-ci | 2025-11-21 09:33:12.671+0000 [id=1] INFO o.e.j.e
jenkins-ci | 2025-11-21 09:33:12.708+0000 [id=1] INFO o.e.j.s
jenkins-ci | 2025-11-21 09:33:12.919+0000 [id=1] INFO hudson.V
jenkins-ci | 2025-11-21 09:33:12.991+0000 [id=1] INFO o.e.j.s
ontextHandler$CoreContextHandler$CoreToNestedHandler@572e6fd9{STARTED}}
INFO
```

Step 4: List docker images

```

(singh@singh)-[~/pr5]
$ sudo docker container ls -a
CONTAINER ID        IMAGE               COMMAND                  CREATED              STATUS              PORTS
ca0bfa73393        pr5-jenkins        "/usr/bin/tini -- /u..." Less than a second ago Up Less than a second ago
802603091e9        program4            "python app.py"         2 weeks ago         Exited (0) 2 weeks ago
eb15eb68d40        3c45199ff050       "flask app.py"          2 weeks ago         Exited (2) 2 weeks ago
46c2302f9c7        program-4           "flask app.py"          2 weeks ago         Exited (2) 2 weeks ago
1dcf51ecd4e        57f1863e1d53       "flask run"             2 weeks ago         Exited (0) 2 weeks ago
f392ffdf571        program3            "python app.py"         2 weeks ago         Exited (0) 2 weeks ago
fa1bc3b7841        52d6d6f80aea       "python app.py"         2 weeks ago         Exited (0) 2 weeks ago
0c42265eda5        52d6d6f80aea       "python app.py"         2 weeks ago         Exited (0) 2 weeks ago
8a518be1f50        7a5ec8552c52       "python app.py"         2 weeks ago         Exited (0) 2 weeks ago
e8fdd8fce8f        5cdc8ed4f59b       "docker-entrypoint.s..." 2 weeks ago         Exited (0) 2 weeks ago
8481cea2cbc        5cdc8ed4f59b       "docker-entrypoint.s..." 2 weeks ago         Exited (0) 2 weeks ago

(singh@singh)-[~/pr5]
$ sudo docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
pr5-jenkins         latest             5ddf84919491       Less than a second ago 925MB
program4            latest             9726101fc3e2       2 weeks ago        1.13GB
program-4           latest             3c45199ff050       2 weeks ago        1.13GB
none>               <none>             57f1863e1d53       2 weeks ago        1.13GB
program3            latest             6ce211a683f4       2 weeks ago        1.13GB

(singh@singh)-[~/pr5]
$ sudo docker volume ls
DRIVER              VOLUME NAME
local               4a2c0f2091cc8e5f56d561acdedc8763f1f04d7091587d99ce740334455ead12
local               5e436ee566bcc18072c46682496186b5bff241304ca00b43fde82f4f18a02d53
local               pr5_jenkins_home

```

Step 5 :Open localhost:8080

localhost:8080

Getting Started

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.